

Digital Clinic Management System

Project Summary

We aim to develop a comprehensive healthcare application that improves the experience for doctors. At its core, the application will feature a user-friendly appointment and billing management system for a clinic. It includes supplementary features such as digital prescription generation and health and clinic insights for doctors, as mentioned below.

To enhance efficiency and reduce paper waste, the app will allow access to digitized prescriptions and medical forms, providing patients with easy access to important documents. Additionally, doctors will be able to track the vitals history of patients, allowing them to monitor health trends over time. The system also offers financial insights, including income trends and billing history, to help them better understand and manage their practice performance effectively. Overall, the application addresses key pain points in healthcare delivery through digitization, transparency, and intuitive design.

Description

Traditional clinics often face challenges in maintaining accurate and accessible patient records, managing appointments, handling billing, and tracking vital health information across multiple visits. Doctors often have to deal with handwritten notes, keep track of billing records by hand, or use different systems that don't work together. This wastes time and creates stress for both doctors and patients, which can affect the quality of care.

We propose a clinic management system to bridge this gap and bring essential parts of clinic operations and patient care into a single digital system. Doctors have a clear and simple way to manage their whole practice in one place. They can easily schedule and keep track of upcoming appointments and review their patients' health history. This helps doctors stay organized and provide better focus on patient care rather than paperwork.

Beyond administrative effectiveness, the system provides doctors with actionable insight into the practice. Graph visualization dashboards illustrate trends in patient visits and income over time, enabling doctors to make decisions on staff scheduling or service offering. By connecting scheduling, patient health record keeping, and financial management, our system improves the efficiency with which the doctor's clinic works.

Unlike many existing systems, our system allows multiple doctors in the same clinic to access and share patient records, so important information isn't kept separate, allowing everyone to stay up to date. It also lets doctors create and share medical documents such as prescriptions and referral forms digitally, which reduces paper use and makes record-keeping and sharing easier.

In short, this project does more than just digitize existing processes. It changes the way doctors and patients interact, saving time, reducing paperwork, and creating a more open and data-driven healthcare environment. Our system aims to improve how clinics run by combining convenience, efficiency, and trust.

Usefulness

The digital clinic management system enables doctors and patients to digitally manage and access their consultation data. Doctors can create, schedule, and track appointments, record consultations, and manage their clinic finances within the system. A variety of doctors of different specializations can be handled in the same application. Doctors would have access to a digital timeline of the patient's consultations, diseases, and trends in vitals.

While digital clinic management systems exist online, we are also bringing in some newer features that current systems do not support. These include the ability to onboard multiple doctors for a single clinic and have shared access to the clinic's patient records. Doctors would be able to digitally generate and share medical documents such as referral forms and prescriptions. We are also developing visualizations tailored to the doctors' needs, such as clinic income graphs and patient visit trends, and body vital trends for each patient.

Creative Components

We have outlined the following creative components that we intend to integrate into the system, which could be helpful to doctors using it.

Trend graphs representing various clinic and patient data

Graphs can be plotted to visualize the clinic's income trends month-on-month, as well as the number of patients visiting the clinic. This would help indicate a trajectory to the doctor with respect to his/her financials. Also, per patient, a graph showing the trajectory of each of their vitals over time, based on data recorded from each of their appointments, can help identify trends and issues.

Automatic PDF generation for prescriptions and other medical documents

We intend to automatically create PDFs, which would autofill patient and doctor details using a service such as Puppeteer, which could convert HTML to PDFs. This would allow us to create a PDF of any medical form that the doctor would like to issue to the patient in a few clicks without the need for the doctor to manually enter information.

Realness

For the proposed clinic management system, we have identified two datasets that would be used to seed the database. The datasets are described below:

1. Healthcare Management System Dataset by anouskaabhisikta ([Kaggle](#))

The above dataset contains CSV data about doctors, patients, appointments, and billing information.

- Appointments - **(5 columns, 900 rows)** Capturing appointment status, slot times, patient, and doctor.
- Doctors - **(4 columns, 900 rows)** Capturing doctor name, specialization, and contact information.
- Patients - **(4 columns, 900 rows)** Capturing patient name and contact information.
- Billing - **(4 columns, 1000 rows)** Capturing patient, amount, and items to be billed.
- Medical Procedures - **(3 columns, 950 rows)** Capturing procedure names and associated appointments.

2. Human Vital Sign Dataset by nasirayub2 ([Kaggle](#))

This CSV dataset contains information on body vitals recorded for 200,000 patients (**17 columns, 200,000 rows**). We would be sampling this data and using it as historical data for the patients in the previous dataset to display a visualization of trends.

Functionality

In this section, we will outline the various features and functionality offered on our digital clinic management system.

Login page

- Option for the doctor to sign into the system as an admin or a consultant doctor.

Doctor Dashboard

- A complete list of all appointments scheduled, ongoing, and completed for the current day.
- A doctor can view appointments for a later or previous day using the arrows beside today's date.
- An admin doctor can view appointments for all the doctors present in that clinic.
- Doctors can change the appointment states for a particular appointment, i.e., Waiting -> Active -> Completed.
- Quick actions will be provided against each appointment for performing actions like completing an appointment, deleting an appointment, generating an invoice, viewing its prescription and invoice, etc.
- The dashboard will also allow the doctor to schedule a new appointment for a particular patient on a particular day.
- Clinic income and patient visits trend graphs will be shown to the doctor on the dashboard.

Patient Search

- A quick search box would be located on the dashboard to quickly search for patients already present in the system.

- The results would allow for creating an instant appointment with a particular patient or go to the patient's profile.

Patients List

- A list of all patients in the clinic sorted in ascending order of their names.
- Clicking on each patient would take the doctor to the respective patient's profile.

Patient Profile

- A complete list of all appointments for the current patient across dates.
- Doctors can update appointment states for an appointment and perform quick actions as present on the dashboard for each appointment on the patient profile as well.
- A doctor can also edit the patient's details, such as name, blood group, contact information, etc.
- The patient profile will also allow the doctor to schedule a new appointment for that patient on a particular day.
- The vital trends graph based on past appointment data for that patient will be shown to the doctor on the patient's profile.

Consultation

- A screen that is shown for each new appointment at the start of the session.
- It includes fields for the doctor to fill out for that appointment, such as vitals, diseases, and notes.
- The data is saved against that appointment and is used to generate a PDF prescription, an invoice, and also update the patient's overall vitals.
- A doctor can end the consultation from this screen.

Invoice

- For each appointment, a new invoice would be created, the contents of which can be edited by the doctor on the invoice page.
- The page allows the doctor to add procedures and other particulars to the invoice and their prices, and also mark the invoice as paid.

Template PDFs

- For each medical template, a PDF version of the data would be generated, filling in the patient's details and appointment data.

Triggers

- On the creation of a new appointment, a new invoice is also generated and attached to that appointment.
- On the completion of an appointment, a PDF prescription is generated from that appointment data, and patient vitals from that appointment are updated against the patient.
- On complete payment of an invoice, clinic income data is updated.
- On deletion of an appointment, delete its invoice data also.

UI mockups

UI-mockups created using Figma AI based on the features listed in this document.

Digi Clinic

Dr. Doctor

Dashboard

Patients

Invoices

Analytics

Dashboard

Welcome back, Dr. Doctor

+ New Appointment

Today's Appointments

XX

Active Patients

XXX

Monthly Revenue

\$XXXXX

Pending Invoices

X

Today's Appointments

Wed, September 17, 2025

09:00

Jahnavi J

General Checkup

waiting

View

09:30

Arjun S

Follow-up

active

View

10:00

Grace L

Consultation

completed

View

10:30

Shiv B

Blood Test

waiting

View

11:00

David D

Physical Exam

scheduled

View

Quick Patient Search

Search patients...

Jahnavi J

Last visit: 2 days ago

Arjun S

Last visit: 1 week ago

Grace L

Last visit: 3 days ago

View All Patients

Monthly Income Trend

[Income Chart: Jan \$45k → Jun \$67k]

Patient Visits This Week

[Visits Chart: Mon 12, Tue 15, Wed 8, Thu 18...]

Doctor Dashboard

Digi Clinic

Dr. Doctor

Dashboard

Patients

Invoices

Analytics

Consultation Session

David D - Physical Exam

X Cancel

Complete Consultation

David D

Patient

11:00

Appointment Time

Physical Exam

Type

Vital Signs

Blood Pressure

120/80

Heart Rate (bpm)

72

Temperature (°F)

98.6

Weight (kg)

70

Quick Actions

View Previous Records

View Vital History

Patient Profile

Recent Vitals

Last BP:

121/80

Last Weight:

73 kg

Last HR:

72 bpm

Consultation Details

Symptoms

List of symptoms observed or reported...

Prescriptions

Findings from physical examination...

Generate Prescription

Save as Draft

Complete & Generate Invoice

Consultation

Patients

Manage your patient records

+ Add New Patient

Search patients...

All (6)Active (5)Inactive (1)

J

Jahnavi J

Age: XX

active

Blood:

+1 (555) 123-4567

jahnavi.j@email.com

Last Visit:

Jun 15, 2025

Next Appointment:

Jun 20, 2025

View Profile

A

Arjun S

Age: XX

active

Blood:

+1 (555) 123-4567

arjun.s@email.com

Last Visit:

Jun 15, 2025

Next Appointment:

Jun 20, 2025

View Profile

G

Grace L

Age: XX

active

Blood:

+1 (555) 123-4567

grace.l@email.com

Last Visit:

Jun 15, 2025

Next Appointment:

Jun 20, 2025

View Profile

S

Shiv B

Age: XX

inactive

Blood:

+1 (555) 123-4567

shiv.b@email.com

Last Visit:

Jun 15, 2025

Next Appointment:

Jun 20, 2025

View Profile

D

David D

Age: XX

active

Blood:

+1 (555) 123-4567

david.d@email.com

Last Visit:

Jun 15, 2025

Next Appointment:

Jun 20, 2025

View Profile

L

Lisa A

Age: XX

active

Blood:

+1 (555) 123-4567

lisa.a@email.com

Last Visit:

Jun 15, 2025

Next Appointment:

Jun 20, 2025

View Profile

Patient List

+ Back to Patients

Lisa A

Patient Profile

+ New Appointment

L

Lisa A

Age: XX Blood Group: A+ ID: P6

+1 (555) 123-4567 lisa.a@email.com

Edit Profile

Overview

Appointments

Vitals & Charts

Documents

Total Visits

24

Last BP

121/80

Heart Rate

72 bpm

Weight

73 kg

Recent Appointments

General Checkup

2025-06-15 at 10:00 AM

completed

Follow-up

2025-05-15 at 2:30 PM

completed

Blood Test

2025-04-10 at 11:15 AM

completed

Patient Profile

Invoice Management

Track and manage patient billing

+ Create Invoice

Total Revenue

\$330,000

Pending

\$200

Overdue

\$300

Total Invoices

5

Search Invoices...

Filter Status

Invoice ID	Patient	Date	Type	Amount	Due Date	Status	Actions
INV-001	John Smith	2025-06-17	Consultation	\$150.00	2025-06-17	paid	View
INV-002	Emily Davis	2025-06-16	Follow-up	\$200.00	2025-06-16	pending	View
INV-003	Michael Brown	2025-06-15	Follow-up	\$300.00	2025-06-15	overdue	View
INV-004	Sarah Wilson	2025-06-17	Follow-up	\$125.00	2025-06-17	draft	View
INV-005	David Johnson	2025-06-14	Consultation	\$180.00	2025-06-14	paid	View

Invoices

Project work distribution

Team Member	Tasks
Arjun Sivaraman	- DDL for doctors (users) table - APIs and CRUD for doctors - Login page UI - Set up application-wide authentication and role-based access - Template PDF generation - Appointment-based triggers - APIs and CRUD for consultation
Jahnavi Juluri	- DDL for Appointments table - APIs and CRUD for appointments - UI for doctor dashboard - Patient search functionality for the dashboard - Add appointment functionality. - Clinic income and patient visits trend graphs
Shiv Bhavsar	- DDL for Patients table - APIs and CRUD for patients - UI for patient list and patient profile - Add and delete patient functionality - Patient vitals trend graph
Grace Lin	- DDL for Invoice table - APIs and CRUD for invoices - UI for invoice list and invoice page - Invoice-based triggers - DDL for Consultation table - Consultation page UI